

Computational perspectives on the Classical Decision Problem

Lecture 5: The Gödel Fragment

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- A formula is in **prenex form** if none of its Boolean connectives out-scopes any of its quantifiers, for example:

Every student with a degree is a graduate student

$$\forall x_1 \forall x_2 (\text{student}(x_1) \wedge \text{degree}(x_2) \wedge \text{has}(x_1, x_2) \rightarrow \text{gradStudent}(x_1)).$$

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- It is a standard result that every first-order formula is logically equivalent to one in prenex form. For example, the non-prenex sentence

$$\forall x_1 (\text{student}(x_1) \wedge \exists x_2 (\text{degree}(x_2) \wedge \text{has}(x_1, x_2)) \rightarrow \text{gradStudent}(x_1))$$

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- The **quantifier prefix** of a prenex-form formula is the word over the alphabet $\{\forall, \exists\}$ formed by its quantifiers.

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- For example,
 - $\exists^*\forall^*$ is the class of prefixes having any number of existential quantifiers followed by exactly any number of universal quantifiers.
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- In particular,

$$\forall x_1 \forall x_2 (\text{student}(x_1) \wedge \text{degree}(x_2) \wedge \text{has}(x_1, x_2) \rightarrow \text{gradStudent}(x_1))$$

has quantifier prefix $\forall\forall$, described by $\forall\forall, \exists^*\forall\forall\exists^*, \forall^*, \dots$

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- Hence,

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- We allow individual constants in quantifier-prefix fragments, but not function symbols.

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- In this lecture, we shall focus on the Gödel fragment, and show that $\text{Sat}([\exists^* \forall^2 \exists^*])$ is decidable.

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- With a additional work, we can eliminate **both** individual constants **and** leading existential quantifiers:

Lemma

The (finite) satisfiability problem for $[\exists^ \forall^2 \exists^*]$ is reducible in polynomial time, to the (finite) satisfiability problem for the subfragment of $[\forall^2 \exists^*]$ without individual constants.*

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- Let $\mathfrak{A}$ be a structure interpreting a purely relational signature, and let $n > 0$. Then we may form the structure $\mathfrak{B}$ with domain $A \times \{0, \dots, n-1\}$ by declaring, for any k -ary predicate r , any elements $a_1, \dots, a_k$ from A , and any integers $i_1, \dots, i_k \in \{0, \dots, n-1\}$,

$$\mathfrak{B} \models r[\langle a_1, i_1 \rangle, \dots, \langle a_k, i_k \rangle] \Leftrightarrow \mathfrak{A} \models r[a_1, \dots, a_k].$$

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- If φ is an equality-free sentence of first-order logic over a purely relational signature, and $\mathfrak{A}$ a structure interpreting that signature. Then $\mathfrak{A} \models \varphi$ if and only if $\mathfrak{A} \cdot n \models \varphi$.

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- If $\mathfrak{A}$ is a structure interpreting Σ , and $\bar{a} = a_1, \dots, a_k$ are (distinct) elements of A , then $\text{tp}^{\mathfrak{A}}[\bar{a}]$ denotes the k -type $a_1, \dots, a_k$ in $\mathfrak{A}$, i.e. the unique k -type $\tau(x_1, \dots, x_k)$ such that $\mathfrak{A} \models \tau[a_1, \dots, a_k]$.

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- We can also think of τ as a structure over the domain $\{x_1, \dots, x_k\}$: for any predicate $r \in \Sigma$ with arity m , and any m -tuple $\bar{w}$ over $\{x_1, \dots, x_k\}$, we set $\tau \models r[\bar{w}]$ just in case the positive literal $r(\bar{w})$ is in τ .

- Remember that a k -type τ always features the variables $x_1, \dots, x_k$, and that we can think of it as a (small) structure.
- Let $\bar{w} = w_1, \dots, w_m$ be a word over the alphabet $\{x_1, \dots, x_k\}$. Then $\tau\langle\bar{w}\rangle$ will denote the m -type whose i th variable is a (fresh) copy of w_i in the structure τ .

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 - $\tau\langle x_1, x_1, x_3, x_4, \dots, x_k \rangle$: the literals of τ not featuring x_2 , but duplicate x_1 and call the copy x_2 .

- Let a $[\forall^2\exists^*]$ -sentence φ of the form

$$\forall x_1 \forall x_2 \exists x_3 \cdots \exists x_n \psi$$

be given.

- We try to construct a **certificate** $\mathfrak{C} = (R, S, T)$, where:
 - R is a set of 1-types;
 - S a set of 2-types; and
 - T a set of n -types consistent with ψ .

We shall insist that $\mathfrak{C}$ posses a collection of properties (listed on the next slide).

- We shall show that φ is satisfiable if and only if such a certificate exists.

- Here are the properties which $\mathfrak{C} = (R, S, T)$ must satisfy:
 - C1 for every $\rho_1, \rho_2 \in R$, there exists $\sigma \in S$ such that $\rho_1 = \sigma \langle x_1 \rangle$ and $\rho_2 = \sigma \langle x_2 \rangle$;
 - C2 for every $\sigma \in S$ there exists $\tau \in T$ such that $\sigma = \tau \langle x_1, x_2 \rangle$;
 - C3 for every $\rho \in R$, there exists $\tau \in T$ such that $\rho = \tau \langle x_1 \rangle$ and $\tau = \tau \langle x_1, x_1, x_3, \dots, x_n \rangle$;
 - C4 for every $\tau \in T$ and every $(1 \leq i \leq n)$, $\tau \langle x_i \rangle \in R$;
 - C5 for every $\tau \in T$ and every $(1 \leq i, j \leq n)$, $\tau \langle x_i, x_j \rangle \in S$.
- The easiest way to motivate these properties is to show that, if φ is satisfiable, then they can always be satisfied.

Lemma

If φ is satisfiable, it has a certificate.

Suppose $\mathfrak{A} \models \varphi$, and let $\mathfrak{B}$ be the Cartesian product $\mathfrak{B} = \mathfrak{A} \cdot n$.
Thus, $\mathfrak{B} \models \varphi$. (Recall that $\varphi := \forall x_1 \forall x_2 \exists x_3 \cdots x_n \psi$.)

R be the set of 1-types realized in $\mathfrak{B}$;

S be the set of 2-types realized in $\mathfrak{B}$; and

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We check that $\mathfrak{C} = (R, S, T)$ satisfies the conditions **C1–C6**.

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C1 for every $\rho_1, \rho_2 \in R$, there exists $\sigma \in S$ such that $\rho_1 = \sigma \langle x_1 \rangle$
and $\rho_2 = \sigma \langle x_2 \rangle$;

If $\rho_1, \rho_2 \in R$, choose **distinct** elements b_1, b_2 realizing these 1-types in $\mathfrak{B}$. Then $\sigma = \text{tp}^{\mathfrak{B}}[b_1, b_2] \in S$ suffices.

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C2 for every $\sigma \in S$ there exists $\tau \in T$ such that $\sigma = \tau \langle x_1, x_2 \rangle$.

If $\sigma \in S$, choose (distinct) elements b_1, b_2 realizing σ in $\mathfrak{B}$. Pick $b_3, \dots, b_n$ in any way such that the elements $\bar{b} = b_1, \dots, b_n$ are **distinct** and $\mathfrak{B} \models \psi[\bar{b}]$. Let $\tau = \text{tp}^{\mathfrak{B}}[\bar{b}]$. $\square$

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$$\pi(x, y) = \frac{1}{2}(x + y - 2)(x + y - 1) + y.$$

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10	14			
6	9	13		
3	5	8	12	
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 $\dots$

- The inverse of the function π maps each $k \in \mathbb{N}^+$ to a pair of positive integers $(\pi_1(k), \pi_2(k))$.

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- Thus, for example $\pi_1(14) = 2$ and $\pi_2(14) = 4$.
- For any pair of positive integers (i, j) the number $k = \pi(i, j)$ satisfies $\pi_1(k) = i$ and $\pi_2(k) = j$.
- Note that $\pi_1(k) \leq k$ and $\pi_2(k) \leq k$.

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- We define a sequence $\{\zeta_k\}_{k \geq 1}$, where ζ_k is a type of arity

$$a(k) = 1 + (k - 1)(n - 2),$$

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- We proceed by induction on k , maintaining the invariants:

$$\begin{aligned} \zeta_k \langle x_1, \dots, x_{a(k-1)} \rangle &= \zeta_{k-1} && \text{if } k > 1; \\ \zeta_k \models \psi(x_{\pi_1(k)}, x_{\pi_2(k)}, x_{a(k-1)+1}, \dots, x_{a(k)}) &&& \text{if } k > 1; \\ \zeta_k \langle x_i \rangle &\in R && \text{for } 1 \leq i \leq a(k); \\ \zeta_k \langle x_i, x_j \rangle &\in S && \text{for all } 1 \leq i, j \leq a(k). \end{aligned}$$

- For $k = 1$, pick any $\rho \in R$ and set $\zeta_1 = \rho$.
- A quick check (using the properties of $\mathfrak{C}$) shows that:

$$\begin{aligned}
 \zeta_k \langle x_1, \dots, x_{a(k-1)} \rangle &= \zeta_{k-1} && \text{if } k > 1; \\
 \zeta_k \models \psi(x_{\pi_1(k)}, x_{\pi_2(k)}, x_{a(k-1)+1}, \dots, x_{a(k)}) &&& \text{if } k > 1; \\
 \zeta_k \langle x_i \rangle &\in R && \text{for } 1 \leq i \leq a(k); \\
 \zeta_k \langle x_i, x_j \rangle &\in S && \text{for all } 1 \leq i, j \leq a(k).
 \end{aligned}$$

The final invariant holds because $\mathfrak{C}$ satisfies C2 and C3, which together guarantee that $\rho \langle x_1, x_1 \rangle \in S$.

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We identify the indices $i_k = \pi_1(k)$ and $j_k = \pi_2(k)$.

Thus $i_k \leq a(k)$ and $j_k \leq a(k)$.

Suppose $i_k \neq j_k$. (The case where $i_k = j_k$ is similar.)

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By IH, $\sigma \in S$.

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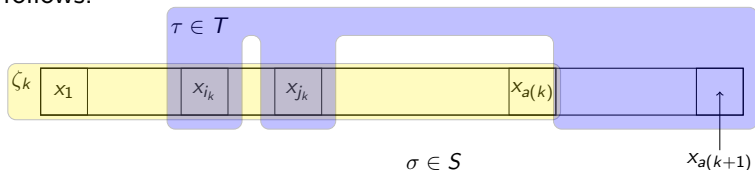
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By **C2**, for every $\sigma \in S$ there exists $\tau \in T$ such that $\sigma = \tau \langle x_1, x_2 \rangle$.

We take $n-2$ new elements,

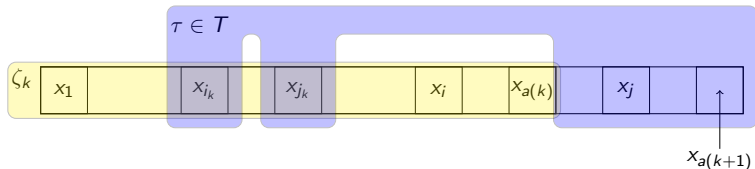
$$x_{a(k)+1}, \dots, x_{a(k)+(n-2)},$$

and assign $x_{i_k}, x_{j_k}, x_{a(k)+1}, \dots, x_{a(k)+(n-2)}$ the n -type τ .

This gives us $a(k)+(n-2) = a(k+1)$ elements altogether—the right number for ζ_{k+1} .

- Suppose then ζ_k has been defined; we construct ζ_{k+1} as follows:

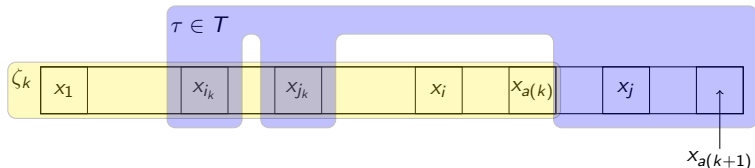
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We now have to complete the n -type ζ_{k+1} . We begin with the 2-types.

Pick distinct indices i and j ; for illustration, assume $1 \leq i \leq a(k)$ and $a(k) < j \leq a(k+1)$.

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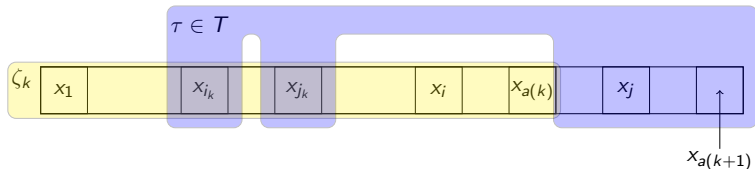


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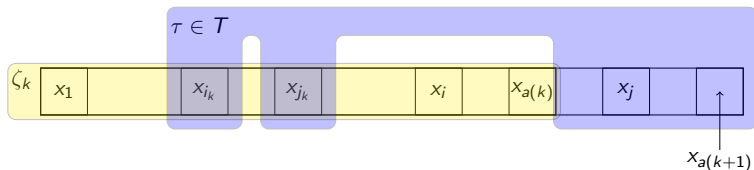
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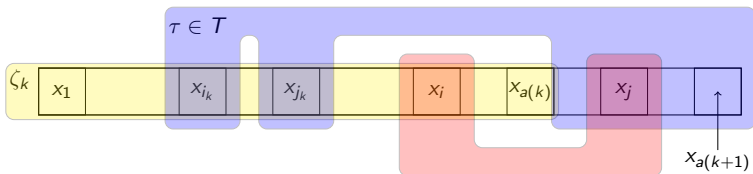
Let ρ_2 be the 1-type assigned to x_j by τ . (By **C4**, $\rho_2 \in R$.)

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By **C1**, we can find a $\sigma \in S$ such that $\rho_1 = \sigma \langle x_1 \rangle$ and $\rho_2 = \sigma \langle x_2 \rangle$.

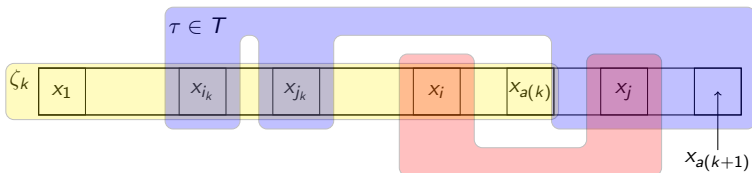
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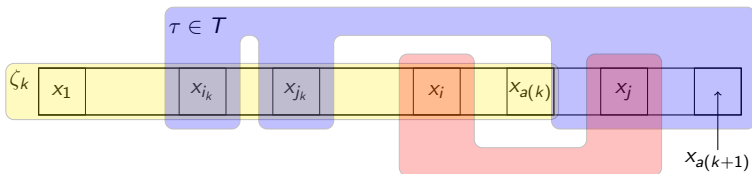


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It is routine to check the required invariants hold for ζ_{k+1} .

- Each $a(k)$ -type ζ_k may be regarded as a structure over $\{x_1, \dots, x_{a(k)}\}$.
- Remembering the invariant

$$\zeta_k \langle x_1, \dots, x_{a(k-1)} \rangle = \zeta_{k-1} \quad \text{if } k > 1,$$

we may define the structure $\mathfrak{X}$, over the domain $X = \{x_1, x_2, x_3, \dots\}$, to be the union of all the ζ_k .

- To see that

$$\mathfrak{X} \models \varphi,$$

remember that $\varphi := \forall x_1 \forall x_2 \exists x_3 \cdots \exists x_n \psi$, and notice that the elements $x_{a(k)+1}, \dots, x_{a(k+1)}$ serve as witnesses for the pair $x_{\pi_1(k)}, x_{\pi_2(k)}$:

$$\mathfrak{X} \models \psi[x_{\pi_1(k)}, x_{\pi_2(k)}, x_{a(k)+1}, \dots, x_{a(k+1)}].$$

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- When considering n -types, we do not care about literals that are not substitution instances of literals in φ .
- It can thus be shown that $\mathcal{C}$ can be written using a number of bits (singly) exponential in $\|\varphi\|$. Hence $\text{Sat}([\exists^*\forall^2\exists^*])$ is in NEXPTIME .

- Finally, recall from Lecture 3 that, if φ is a sentence of $\mathcal{FO}^2$, then we can compute, in polynomial time, a sentence ψ of the form

$$\forall x \forall y \alpha(x, y) \wedge \bigwedge_{h=1}^m \forall x \exists y \beta_h(x, y),$$

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- This gives us ...

Theorem

The problem $\text{Sat}([\exists^ \forall^2 \exists^*])$ is NEXPTIME-complete.*

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- An elegant **probabilistic** proof for the FMP for $[\exists^* \forall^2 \exists^*]$ was given by Y. Gurevich and S. Shelah in 1983.
- In any case, the proof depends essentially on the absence of equality. In 1985, W. Goldfarb showed that $\text{Sat}([\exists^* \forall^2 \exists^*]_=)$ and $\text{FinSat}([\exists^* \forall^2 \exists^*]_=)$ are both undecidable.

- The last sentences of Gödel's 1933 paper contains the sentence

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- A correct proof (including equality) was given by M. Mortimer in 1975.